

Probabilistic Combinatorics

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1. Extremal Combinatorics

Lecture 1

Question 1.0.1. How large can a structure be if it satisfies certain properties?

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Example 1.0.2 (Mantel, 1907). How many edges can a graph in n vertices have if it doesn't contain a triangle?

To show the answer is $f(n)$, we need to do two things.

- (1) Show there is a triangle-free graph on n vertices with $f(n)$ edges.
- (2) Prove that any n -vertex graph with $> f(n)$ edges must have a triangle.

Theorem 1.0.3 (Mantel, 1907)

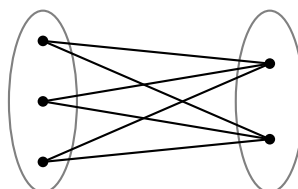
The largest n -vertex triangle-free graph has $\left\lfloor \frac{n^2}{4} \right\rfloor$ edges.

Remark 1.0.4. The complete graph K_n on n vertices has $\binom{n}{2} \approx \frac{1}{2}n^2$ edges.

proof of Mantel's theorem.

- (1) Bipartite graphs are triangle-free. Thus, we can split the vertices into two parts, take all edges between the parts. Suppose first part is of a vertices and the other one is of $n - a$ vertices.

$A, |A| = a \qquad B, |B| = n - a$



$K_{a, n-a}$: all edges between the two parts

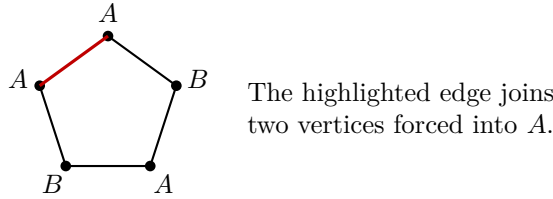
Note that this construction is triangle-free because given any three vertices, two fall in same part, and thus there is no edge between them. Note that the number of edges is $a(n - a) \leq \left\lfloor \frac{n^2}{4} \right\rfloor$, where the equality holds iff $a = \left\lfloor \frac{n}{2} \right\rfloor$ or $\left\lceil \frac{n}{2} \right\rceil$.

Note 1.0.5. Every bipartite graph is triangle-free, but a triangle-free graph need not be bipartite. For example, C_5 consists of the five edges of a pentagon, with no diagonals, so it contains no triangle.

To see why C_5 is not bipartite, suppose that its vertices could be partitioned into two sets A and B , with every edge joining different sets. Put the top vertex in A ; we can always exchange the names of the two sets to do this. Moving clockwise around the pentagon, each vertex must belong to the opposite set from the previous vertex. Thus the five vertices are forced to belong to

$$A, \quad B, \quad A, \quad B, \quad A.$$

But the last vertex is adjacent to the first vertex, and both are in A , a contradiction. Hence no bipartition exists.

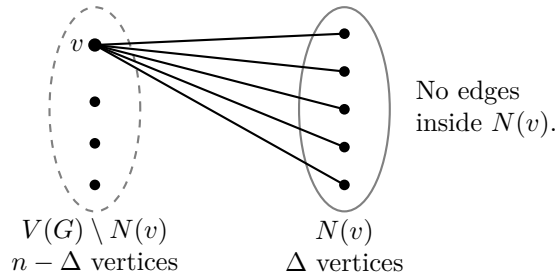


The labels A, B indicate the proposed parts, not vertex names. The red edge is an ordinary edge of C_5 , highlighted only to show the contradiction; this is not an edge-coloring problem.

(2)

Claim 1.0.1. If G is an n -vertex, triangle-free graph, then $e(G) \leq \left\lfloor \frac{n^2}{4} \right\rfloor$.

proof of claim. Let $v \in V(G)$ be a vertex of maximum degree Δ . Since G is triangle-free, the neighborhood of v is independent (no edges).



Thus, every edge contains at least one vertex of $V(G) \setminus N(v)$, where this set includes v itself. Thus,

$$e(G) \leq \sum_{u: u \sim v} \# \{ \text{edges containing } u \} = \sum_{u: u \sim v} \deg(u) \leq \sum_{u: u \sim v} \Delta = \Delta(n - \Delta) \leq \left\lfloor \frac{n^2}{4} \right\rfloor.$$

■

Remark 1.0.6. Up to isomorphism, the unique n -vertex triangle-free graph with $\lfloor n^2/4 \rfloor$ edges is the balanced complete bipartite graph

$$K_{\lfloor n/2 \rfloor, \lceil n/2 \rceil}.$$

To prove uniqueness, suppose that G attains this bound. As in the proof above, choose a vertex v of maximum degree Δ , and write

$$A = N(v), \quad B = V(G) \setminus N(v).$$

Then $|A| = \Delta$, $|B| = n - \Delta$, and A is independent because G is triangle-free. The inequalities in the proof become

$$\left\lfloor \frac{n^2}{4} \right\rfloor = e(G) \leq \sum_{u \in B} \deg(u) \leq \Delta(n - \Delta) \leq \left\lfloor \frac{n^2}{4} \right\rfloor.$$

Since the first and last quantities agree, every inequality must be an equality. Each equality gives a structural restriction:

- (i) **There are no edges inside B .** Let $e(B)$ denote the number of edges with both endpoints in B . Since there are no edges inside A , the degree sum over B counts every edge between A and B once and every edge inside B twice. Thus

$$\sum_{u \in B} \deg(u) = e(G) + e(B).$$

Equality $e(G) = \sum_{u \in B} \deg(u)$ forces $e(B) = 0$. Both A and B are therefore independent, so G is bipartite with these parts.

- (ii) **Every possible edge between the parts is present.** Each $u \in B$ has $\deg(u) \leq \Delta$. Equality

$$\sum_{u \in B} \deg(u) = |B|\Delta$$

forces $\deg(u) = \Delta$ for every $u \in B$; otherwise the sum would be strictly smaller. All neighbors of u lie in A , which has exactly Δ vertices. Hence every vertex of B is adjacent to every vertex of A , giving $G = K_{\Delta, n-\Delta}$.

- (iii) **The two parts are as equal in size as possible.** Finally, $\Delta(n - \Delta) = \lfloor n^2/4 \rfloor$. Using

$$\Delta(n - \Delta) = \frac{n^2 - (2\Delta - n)^2}{4},$$

we obtain $2\Delta - n = 0$ when n is even, and $2\Delta - n = \pm 1$ when n is odd. Thus the part sizes are $\lfloor n/2 \rfloor$ and $\lceil n/2 \rceil$.

Conversely, this balanced complete bipartite graph is triangle-free and has exactly $\lfloor n^2/4 \rfloor$ edges, as shown in the construction. This proves both extremality and uniqueness up to relabeling vertices. ■

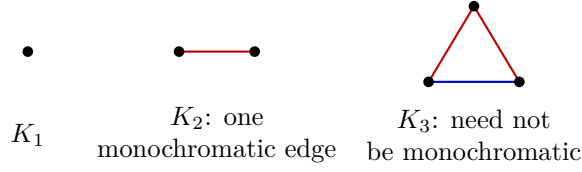
Definition 1.0.7. The **Ramsey number** $R(k)$ is the smallest n s.t. every red-/blue-coloring of the edges of K_n contains a monochromatic copy of K_k .

Theorem 1.0.8 (Ramsey, 1930)

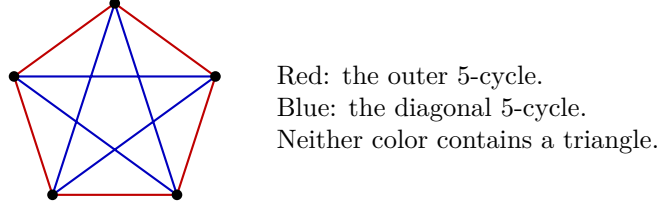
$R(k)$ is finite.

We can simply show $R(1) = 1$, $R(2) = 2$, and for $R(3)$, it is a little harder to show $R(3) = 6$.

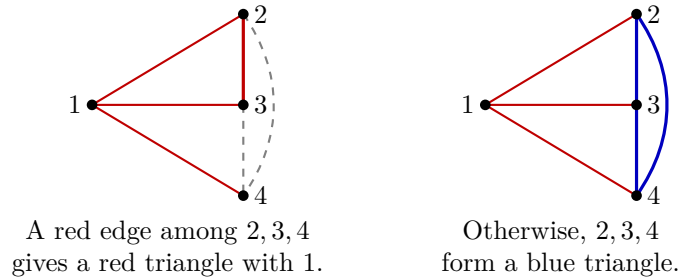
Claim 1.0.2. $R(1) = 1$, $R(2) = 2$, and $R(3) = 6$.



We can first show $R(3) > 5$ by giving an example:



We can also show $R(3) \leq 6$ by pigeonhole principle: Consider K_6 with vertices indexed by 1, 2, 3, 4, 5, 6, we know there exists 3 edges incident to vertex 1 are of same colors, WLOG, say vertices 2, 3, 4 are of red. Then we can show that no matter how we color $\{2, 3\}$, $\{3, 4\}$, $\{2, 4\}$, there must be a monochromatic triangle.



Only the relevant edges are shown; vertices 5, 6 and all other edges can be ignored. Hence every red-/blue-coloring of K_6 has a monochromatic triangle, so $R(3) = 6$.

Theorem 1.0.9 (Erdos-Szekeres, 1935)

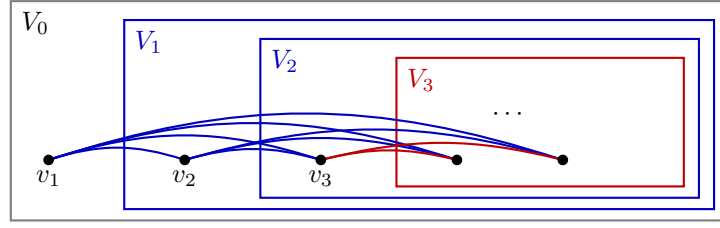
$$R(k) \leq 4^k.$$

Proof. Let $k \geq 2$, and consider any red-/blue-coloring of $K_{2^{2k}}$. Set $V_0 = V(K_{2^{2k}})$. We successively choose vertices $v_1, v_2, \dots$, colors $c_1, c_2, \dots$, and nested sets $V_0 \supseteq V_1 \supseteq V_2 \supseteq \dots$.

At step i , choose $v_i \in V_{i-1}$. Among the edges from v_i to $V_{i-1} \setminus \{v_i\}$, at least half have the same color, say c_i . Let V_i be the set of endpoints of these edges. Then every edge from v_i to V_i has color c_i , and

$$|V_i| \geq \left\lceil \frac{|V_{i-1}| - 1}{2} \right\rceil \geq 2^{2k-i}.$$

The last inequality follows inductively from $|V_0| = 2^{2k}$; the ceiling accounts for removing v_i before keeping a color class.



Keep a monochromatic neighborhood at each step.

In particular, for $j < \ell \leq i$, we have $v_\ell \in V_j$, so the edge $v_j v_\ell$ has color c_j . Also, every edge from v_j to V_i has color c_j .

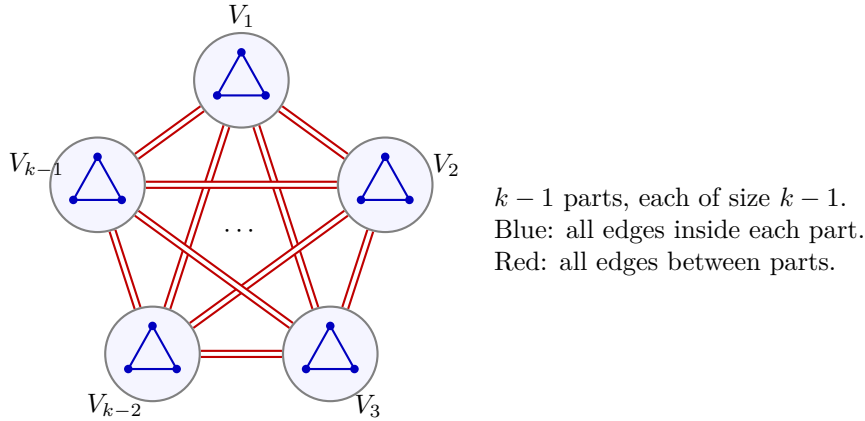
Repeat until $i = 2k$. This is possible since $|V_{2k-1}| \geq 2$, and hence V_{2k} still contains at least one vertex. We obtain

vertex	v_1	v_2	v_3	$\cdots$	v_{2k-1}	v_{2k}
color	c_1	c_2	c_3	$\cdots$	c_{2k-1}	c_{2k}

By the pigeonhole principle, at least k of the colors $c_1, \dots, c_{2k}$ are equal. Choose indices $i_1 < \dots < i_k$ with this common color. For any $r < s$, the edge $v_{i_r} v_{i_s}$ has color c_{i_r} , so these k vertices form a monochromatic K_k . Thus $R(k) \leq 2^{2k} = 4^k$. The case $k = 1$ is immediate. ■

Remark 1.0.10. More careful counting gives $R(k) \leq \binom{2k-2}{k-1}$.

For the lower bound, we can use a smart construction to show $R(k) > (k-1)^2$: For $k \geq 2$, partition the vertices of $K_{(k-1)^2}$ into $k-1$ parts $V_1, \dots, V_{k-1}$, each of size $k-1$. Color all edges inside a part blue and all edges between different parts red.



The diagram is schematic: each circle represents an entire blue K_{k-1} , and each red connection represents all edges between two parts. A blue clique must lie in a single part, so it has at most $k-1$ vertices. A red clique contains at most one vertex from each part, so it also has at most $k-1$ vertices. Thus this coloring has no monochromatic K_k , proving $R(k) > (k-1)^2$.

Theorem 1.0.11 (Erdos, 1947)

$$R(k) > (\sqrt{2} - o(1))^k = 2^{(\frac{1}{2} - o(1))k}.$$

Proof. Take the complete graph K_n on n vertices. Coloring each edge red or blue with probability $\frac{1}{2}$, independently.

The bad events, say E_S , is the event that set S of k vertices whose edges are all the same color. Then

$$\mathbb{P}(E_S) = 2 \cdot \left(\frac{1}{2}\right)^{\binom{k}{2}}.$$

Then

$$\begin{aligned}\mathbb{P}(\text{coloring is bad}) &= \mathbb{P}\left(\bigcup_{S \in \binom{V}{k}} E_S\right) \leq \sum_{S \in \binom{V}{k}} \mathbb{P}(E_S) = \binom{n}{k} 2 \left(\frac{1}{2}\right)^{\binom{k}{2}} < 2n^k 2^{-\frac{k(k-1)}{2}} \\ &= 2 \left(\frac{n}{2^{\frac{k-1}{2}}}\right)^k.\end{aligned}$$

If $n < (1-c)2^{\frac{k-1}{2}}$ for some constant $c > 0$, then

$$\mathbb{P}(\text{bad coloring}) \leq 2(1-c)^k \rightarrow 0,$$

and thus

$$\mathbb{P}(\text{coloring has no monochromatic } K_k) > 0,$$

i.e. $\exists$ a coloring of K_n without monochromatic K_k . Hence, we must have $R(k) > n$. ■

Theorem 1.0.12

$$2^{\frac{1}{2}k} \lesssim R(k) \leq 2^{2k}.$$

Definition 1.0.13 (Extremal problem). An **extremal problem** is a problem to find the size of a largest structure with a desired property.

To solve such problem, we should

- (1) show a structure of large size with given property. (lower bound)
- (2) prove that larger structures can't have property. (upper bound)

Definition 1.0.14 (Probabilistic method). To use **probabilistic method** to solve a problem, you should construct a probability space on the set of structures, then show that the desired property has positive parts, which shows the existence of a structure with such property.

Lecture 2

Recall . Erdős's lower bound on $R(k)$ colored the edges of K_n red and blue uniformly at random and applied the union bound over all k -sets of vertices: if

$$\binom{n}{k} 2^{1-\binom{k}{2}} < 1,$$

then with positive probability no copy of K_k is monochromatic, so $R(k) > n$.

That argument never used that the colored objects are the edges of a complete graph. All it needed was a family of sets, the edge sets of the copies of K_k , none of which may end up monochromatic. This lecture studies that coloring problem in its own right.

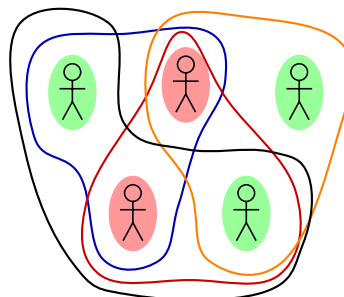
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§1.1. Property B

§1.1.i. Two-Colorability of Hypergraphs

Example 1.1.1 (Chess Olympiad). A national team travels to the Chess Olympiad. Its members fill the following roles.

- Team manager
- Team captains
- Women
- Endgame specialists
- Opening experts
- e4 players
- d4 players
- Sport psychologist



A member can hold several roles, e.g. a team captain who is also an endgame specialist, and a role is usually held by several members. The e4 and d4 players are those whose usual first move is 1.e4 and 1.d4, respectively. The team travels on two flights, and each flight should carry someone in every role, so that the team is complete even if the other flight is delayed.

Question 1.1.2. How can we assign the members to the 2 flights so that each flight has a person with every role?

The picture shows five members and the four underlined roles. Each closed curve encloses the members holding the role underlined in the same color, and the shading behind each member is a flight assignment: the members shaded red take one flight, those shaded green take the other. This answers the question, since each of the four curves contains both a red and a green member.

Abstractly, the data are a set of people with a family of subsets of it, one subset per role, and a flight assignment colors the people with two colors. This is the following structure.

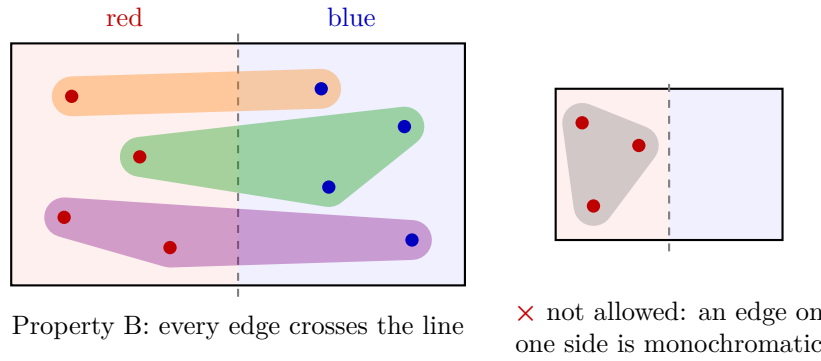
Definition 1.1.3 (Hypergraph). A **hypergraph** is a pair $H = (V, E)$ of vertices V and edges $E \subseteq 2^V$, where an edge is a subset of vertices. It is **k -uniform** if $|e| = k$ for all $e \in E$.

Notation . 2^V is the power set of V , and $V(H)$, $E(H)$ are the vertex and edge sets of H ; all hypergraphs here are finite. For a set X , $\binom{X}{k}$ is the family of all k -element subsets of X , and $[N] = \{1, \dots, N\}$.

Remark 1.1.4. A graph is a 2-uniform hypergraph: each of its edges is a set $\{u, v\}$ of two vertices.

Definition 1.1.5 (Property B). A hypergraph has **Property B**, or is **two-colorable**, if we can two-color the vertices of it s.t. every edge has vertices of both 2 colors.

Picture the two colors as the two halves of a box: put the red vertices in the left half and the blue vertices in the right half. Property B asks for a way to place every vertex so that each edge crosses the dividing line, i.e. has a vertex on both sides.



Equivalently, there is a coloring $\chi: V \rightarrow \{\text{red}, \text{blue}\}$ under which no edge is **monochromatic**, i.e. has all of its vertices of one color. The name refers to Felix Bernstein, who studied this property in 1908.

In [Example 1.1.1](#), the vertices are the members, each role is an edge, and a flight assignment is a two-coloring, so the question asks whether this hypergraph has Property B. The shading shows that the drawn part, a 3-uniform hypergraph with five vertices and four edges, does.

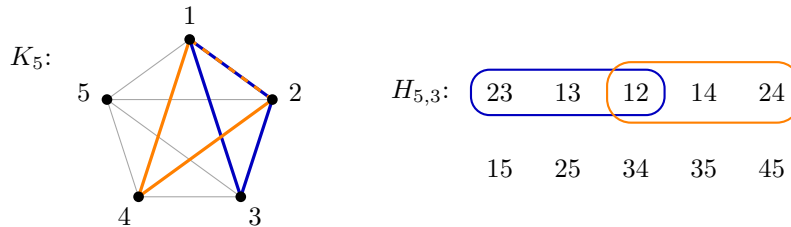
§1.1.ii. Connection to Ramsey Numbers

The Ramsey problem is a question about Property B for one particular family of hypergraphs. Given n and k , build the hypergraph $H_{n,k}$ with

$$V(H_{n,k}) = E(K_n), \quad E(H_{n,k}) = \{E(K) : K \text{ is a copy of } K_k \text{ in } K_n\}.$$

Its vertices are the $\binom{n}{2}$ edges of K_n . It has one edge for each k -set S of vertices of K_n , consisting of the $\binom{k}{2}$ edges of K_n inside S . Thus $H_{n,k}$ is a $\binom{k}{2}$ -uniform hypergraph with $\binom{n}{k}$ edges.

Example 1.1.6 ($n = 5, k = 3$). Write ij for the edge of K_5 between the vertices i and j .



The ten vertices of $H_{5,3}$ are the ten edges of K_5 , and its ten edges are the triangles of K_5 , each a set of three edges of K_5 . Only two triangles are drawn. The blue triangle 123 is the edge $\{12, 13, 23\}$ of $H_{5,3}$, and the orange triangle 124 is the edge $\{12, 14, 24\}$. These two edges of $H_{5,3}$ share the vertex 12 because the two triangles share the edge 12, drawn in both colors. The colors only mark the two triangles; they are not an edge-coloring.

A red/blue coloring of the edges of K_n is the same thing as a two-coloring of the vertices of $H_{n,k}$, and a copy of K_k is monochromatic exactly when the corresponding edge of $H_{n,k}$ is. Hence K_n has a coloring without a monochromatic K_k iff $H_{n,k}$ has Property B, and

$$R(k) = \min\{n : H_{n,k} \text{ does not have Property B}\}.$$

Unfolding the definition of Property B, $H_{n,k}$ is without Property B iff every two-coloring of its vertices leaves some edge monochromatic, so

$$R(k) = \min\{n : \forall \chi: V(H_{n,k}) \rightarrow \{\text{red}, \text{blue}\}, \exists e \in E(H_{n,k}) \text{ monochromatic under } \chi\}.$$

Since $V(H_{n,k}) = E(K_n)$, such a χ is a red/blue coloring of the edges of K_n , and a monochromatic $e \in E(H_{n,k})$ is a monochromatic copy of K_k ; this is word for word the definition of $R(k)$ from Lecture 1.

Remark 1.1.7. For 2-uniform hypergraphs, Property B $\equiv$ bipartite: a 2-uniform hypergraph is a graph, and an edge $\{u, v\}$ sees both colors iff u and v get different colors. So a two-coloring without monochromatic edges is exactly a partition of V into two independent sets, i.e. a bipartition. Property B thus generalizes bipartiteness to hypergraphs; for instance, C_5 from Lecture 1, and every other odd cycle, does not have Property B.

§1.1.iii. The Extremal Problem

Every edge is one more constraint on the coloring, so a hypergraph with few edges should be two-colorable. We measure the size of a hypergraph by its number of edges and leave the number of vertices unrestricted.

Question 1.1.8 (Extremal problem). How small can a k -uniform hypergraph be if it does not have Property B?

Definition 1.1.9.

$$m(k) := \min\{m : \exists k\text{-uniform hypergraph } H = (V, E), |E| = m, \text{ without Property B}\}.$$

Unfolding the definition of Property B, H is without Property B iff every two-coloring of V leaves some edge monochromatic, so

$$m(k) = \min \left\{ m : \begin{array}{l} \exists H = (V, E) \text{ with } |E| = m \text{ and } |e| = k \ \forall e \in E, \text{ s.t.} \\ \forall \chi: V \rightarrow \{\text{red, blue}\}, \exists e \in E \text{ monochromatic under } \chi \end{array} \right\}.$$

Here $\min \emptyset = \infty$, so $m(k) = \infty$ would mean that every k -uniform hypergraph has Property B; [Corollary 1.1.13](#) shows this does not happen. By minimality, every k -uniform hypergraph with fewer than $m(k)$ edges has Property B, while some k -uniform hypergraph with $m(k)$ edges does not.

Proposition 1.1.10 (Erdős, 1963)

$$m(k) \geq 2^{k-1}.$$

Proof. WTS: If $H = (V, E)$ is a k -uniform hypergraph with $m < 2^{k-1}$ edges, then it is two-colorable.

Color the vertices red and blue independently and uniformly. Let $E = \{e_1, e_2, \dots, e_m\}$, and let A_i be the event that the edge e_i is monochromatic. The edge e_i has exactly k vertices with independent colors, and it is monochromatic iff they are all red or all blue, so

$$\mathbb{P}(A_i) = 2 \left(\frac{1}{2} \right)^k = \left(\frac{1}{2} \right)^{k-1}.$$

The events A_i need not be independent, since edges may share vertices, but the union bound does not need independence:

$$\mathbb{P}(\exists \text{ monochromatic edge}) = \mathbb{P}\left(\bigcup_{i=1}^m A_i\right) \leq \sum_{i=1}^m \mathbb{P}(A_i) = m \cdot 2^{-(k-1)} < 1 \quad \text{for } m < 2^{k-1}.$$

Taking complements, and using that an event of positive probability contains at least one

outcome,

$$\begin{aligned}
 & \mathbb{P}(\# \text{ monochromatic edge}) > 0 \\
 & \Downarrow \\
 & \exists \text{ a two-coloring of vertices where every edge sees both colors} \\
 & \Downarrow \\
 & H \text{ is two-colorable.} \quad \blacksquare
 \end{aligned}$$

Remark 1.1.11. Applied to $H_{n,k}$, which is $\binom{k}{2}$ -uniform with $\binom{n}{k}$ edges, [Proposition 1.1.10](#) says: if $\binom{n}{k} < 2^{\binom{k}{2}-1}$, then $H_{n,k}$ has Property B, i.e. $R(k) > n$. This is exactly the condition $\binom{n}{k} 2^{1-\binom{k}{2}} < 1$ in Erdős's lower bound on $R(k)$, and the proof above is the same computation with K_n replaced by an arbitrary hypergraph.

§1.1.iv. Upper Bounds on $m(k)$

An upper bound on $m(k)$ comes from exhibiting one k -uniform hypergraph without Property B that has few edges.

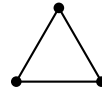
Proposition 1.1.12

$m(1) = 1$, $m(2) = 3$, and for every $k \geq 1$,

$$m(k) \leq \binom{2k-1}{k} \sim c \frac{2^{2k}}{\sqrt{k}}, \quad c = \frac{1}{2\sqrt{\pi}}.$$



$k = 1$: one vertex, one edge



$k = 2$: the triangle K_3

Proof.

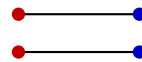
- (1) $m(1) = 1$: Take one vertex v and the single edge $\{v\}$ (left picture above). This edge has only one vertex, so it is monochromatic under every coloring. A hypergraph with no edges has Property B vacuously, so $m(1) = 1$. In [Example 1.1.1](#), an edge of size 1 is a role held by a single member, e.g. a lone sport psychologist: then no flight assignment works.
- (2) $m(2) = 3$: A 2-uniform hypergraph is a graph, and by [Remark 1.1.7](#) it is without Property B iff it is not bipartite. So $m(2)$ is the fewest edges of a non-bipartite graph, and we show two things.
 - $m(2) \leq 3$: The triangle K_3 (right picture above) is without Property B. Take any two-coloring of its three vertices. By the pigeonhole principle, two of the three vertices get the same color, and since every two vertices of K_3 are adjacent, the edge between them is monochromatic.
 - $m(2) \geq 3$: Every graph G with at most two edges has Property B. We color the vertices of the edges red and blue as below, and all other vertices, which lie in no edge, arbitrarily.

$\emptyset$

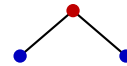
no edges



one edge



two disjoint
edges

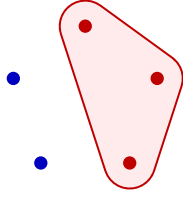


two edges sharing
a vertex

With one edge $\{u, v\}$, color u red and v blue. With two disjoint edges $\{a, b\}$, $\{c, d\}$, color a, c red and b, d blue. With two edges $\{a, b\}$, $\{a, c\}$ sharing the vertex a , color a red and b, c blue. In every case each edge has one red and one blue endpoint, so no edge is monochromatic.

Hence $m(2) = 3$. Note that [Proposition 1.1.10](#) alone only gives $m(2) \geq 2^{2-1} = 2$, so the case check above is needed for the lower bound.

- (3) $m(k) \leq \binom{2k-1}{k}$: Consider $2k - 1$ vertices and let any k of them form an edge, i.e. take the hypergraph $([2k - 1], \binom{[2k-1]}{k})$. In a two-coloring of $2k - 1$ vertices, by the pigeonhole principle, some color is used on at least k vertices, and any k of these form a monochromatic edge.



$k = 3$: five vertices, any three of which form an edge. In this coloring three vertices are red, and they form a monochromatic edge.

So this hypergraph does not have Property B, and $m(k) \leq \binom{2k-1}{k}$. For $k = 1, 2$, it is exactly the edge $\{v\}$ and the triangle above, so the bound is tight there.

- (4) Asymptotics: By Pascal's rule and $\binom{2k-1}{k-1} = \binom{2k-1}{k}$,

$$\binom{2k}{k} = \binom{2k-1}{k-1} + \binom{2k-1}{k} = 2\binom{2k-1}{k},$$

and Stirling's formula $N! \sim \sqrt{2\pi N}(N/e)^N$ gives

$$\binom{2k}{k} = \frac{(2k)!}{(k!)^2} \sim \frac{\sqrt{4\pi k} (2k/e)^{2k}}{2\pi k (k/e)^{2k}} = \frac{4^k}{\sqrt{\pi k}}.$$

Hence $\binom{2k-1}{k} = \frac{1}{2} \binom{2k}{k} \sim \frac{1}{2\sqrt{\pi}} \cdot \frac{2^{2k}}{\sqrt{k}}.$ ■

Corollary 1.1.13

$m(k)$ is finite for every $k \geq 1$.

Proof. The hypergraph $([2k - 1], \binom{[2k-1]}{k})$ in the proof of [Proposition 1.1.12](#) is k -uniform and does not have Property B, so $m(k) \leq \binom{2k-1}{k} < \infty$. ■

Remark 1.1.14. Together, [Proposition 1.1.10](#) and [Proposition 1.1.12](#) give

$$2^{k-1} \leq m(k) \leq (1 + o(1)) \frac{4^k}{2\sqrt{\pi k}}.$$

Both bounds are exponential in k , with bases 2 and 4, the same kind of gap as $\sqrt{2}^k \lesssim R(k) \leq 4^k$ for Ramsey numbers. Unlike for Ramsey numbers, this gap can be closed up to a polynomial factor: the base 2 of the lower bound is the right one. The explicit hypergraph $([2k-1], \binom{[2k-1]}{k})$ is wasteful because it uses every k -set of its vertices, and the next subsection shows that a *random* choice of far fewer k -sets on a larger vertex set already fails to have Property B.

§1.1.v. A Random Construction

To show that $m(k) \leq m$ we must exhibit a k -uniform hypergraph with m edges and without Property B. Instead of constructing one, we pick one at random and show that it fails to have Property B with positive probability.

Theorem 1.1.15 (Erdős, 1964)

$m(k) = O(k^2 2^k)$, i.e. there is a constant $C > 0$ s.t. $m(k) \leq Ck^2 2^k$ for all large enough k .

Proof. Idea: take a random hypergraph H .

- Fix a set V of n vertices.
- Choose m edges $e_1, e_2, \dots, e_m$ uniformly at random from $\binom{V}{k}$, independently of each other.

The numbers n and m will be chosen later. The same k -set may be picked twice; then H has fewer than m distinct edges, and deleting repeated copies of an edge does not affect Property B. So it suffices to show

$$\mathbb{P}(H \text{ is not two-colorable}) > 0 \Rightarrow m(k) \leq |E(H)| \leq m.$$

Now the random object is the hypergraph, and the bad event is that H is two-colorable, i.e. for some two-coloring $\chi: V \rightarrow \{\text{red}, \text{blue}\}$ there is no monochromatic edge. There are 2^n colorings, so we fix one, bound the probability that it works, and take a union bound over all of them. This is the reverse of Proposition 1.1.10, where the hypergraph was fixed and the coloring was random.

Let χ be a two-coloring of V , and let A_χ be the event that none of $e_1, e_2, \dots, e_m$ is monochromatic under χ . Consider one edge e_i . Since e_i is uniformly random,

$$\mathbb{P}(e_i \text{ is monochromatic under } \chi) = \frac{\# \text{ monochromatic } k\text{-sets}}{\# \text{ possible } k\text{-sets}} = \frac{\binom{r}{k} + \binom{n-r}{k}}{\binom{n}{k}},$$

where r is the number of red vertices under χ . The function $x \mapsto \binom{x}{k} = \frac{x(x-1)\cdots(x-k+1)}{k!}$, extended by 0 for $x < k-1$, is convex on $[0, \infty)$.¹ By convexity,

$$\frac{1}{2} \left[\binom{r}{k} + \binom{n-r}{k} \right] \geq \binom{\frac{1}{2}(r+n-r)}{k} \Rightarrow \binom{r}{k} + \binom{n-r}{k} \geq 2 \binom{n/2}{k},$$

that is, a balanced coloring has the fewest monochromatic k -sets. Hence, for every χ ,

$$\mathbb{P}(e_i \text{ is monochromatic under } \chi) \geq \frac{2 \binom{n/2}{k}}{\binom{n}{k}} =: p,$$

where $\binom{n/2}{k}$ is the polynomial above evaluated at $x = n/2$, so n need not be even. The edges are chosen independently, so

$$\begin{aligned} \mathbb{P}(A_\chi) &= \mathbb{P}(\nexists \text{ monochromatic edge under } \chi) = \prod_{i=1}^m (1 - \mathbb{P}(e_i \text{ is monochromatic under } \chi)) \\ &\leq (1 - p)^m, \end{aligned}$$

¹On $[k-1, \infty)$ it is a polynomial all of whose roots are $\leq k-1$, so by Rolle's theorem all of its derivatives are positive there; gluing it to the constant 0 at $x = k-1$, where both vanish and the right derivative is ≥ 0 , keeps it convex.

and by the union bound over the 2^n colorings,

$$\mathbb{P}(H \text{ is two-colorable}) = \mathbb{P}\left(\bigcup_{\chi} A_{\chi}\right) \leq \sum_{\chi} \mathbb{P}(A_{\chi}) \leq 2^n(1-p)^m.$$

We must choose n and m so that $2^n(1-p)^m < 1$, where $p = 2^{\binom{n/2}{k}}/\binom{n}{k}$. Using $1-x < e^{-x}$ for $x > 0$,

$$2^n(1-p)^m < 2^n e^{-pm} = e^{n \ln 2 - pm}, \quad \text{where } e^{n \ln 2 - pm} \leq 1 \Leftrightarrow m \geq \frac{n \ln 2}{p}.$$

We will take $m = \lfloor \frac{n \ln 2}{p} \rfloor + 1$, and choose n to minimize it. For this we need p precisely as a function of n and k . Writing out both binomial coefficients, the $i = 0$ factor equals $\frac{1}{2}$ and cancels the 2:

$$\begin{aligned} p &= \frac{2^{\binom{n/2}{k}}}{\binom{n}{k}} = 2 \prod_{i=0}^{k-1} \frac{\frac{n}{2} - i}{n - i} = \prod_{i=1}^{k-1} \frac{n - 2i}{2(n - i)} = \prod_{i=1}^{k-1} \left(\frac{1}{2} - \frac{i}{2(n - i)} \right) = \frac{1}{2^{k-1}} \prod_{i=1}^{k-1} \left(1 - \frac{i}{n - i} \right) \\ &= \frac{1}{2^{k-1}} \prod_{i=1}^{k-1} \left(1 - \frac{i}{n} + O\left(\frac{i^2}{n^2}\right) \right) = \frac{1}{2^{k-1}} \prod_{i=1}^{k-1} e^{-\frac{i}{n} + O\left(\frac{i^2}{n^2}\right)} \\ &= \frac{1}{2^{k-1}} e^{-\sum_{i=1}^{k-1} \left(\frac{i}{n} + O\left(\frac{i^2}{n^2}\right) \right)} = \frac{1}{2^{k-1}} e^{-\frac{k(k-1)}{2n} + O\left(\frac{k^3}{n^2}\right)}. \end{aligned}$$

Here we assumed $n \geq 2k$, so that $\frac{i}{n-i} \leq 1$ and $\frac{i}{n-i} = \frac{i}{n} + O\left(\frac{i^2}{n^2}\right)$, and used $1-x = e^{-x+O(x^2)}$ for small $x \geq 0$. Now, since

$$m \approx \frac{n \ln 2}{p} = 2^{k-1} n e^{\frac{k(k-1)}{2n} + O\left(\frac{k^3}{n^2}\right)} \ln 2,$$

we minimize $n e^{k^2/(2n)}$ over n : its derivative in n is $e^{k^2/(2n)}(1 - \frac{k^2}{2n})$, which vanishes at $n = \frac{k^2}{2}$. The optimal choice is thus $n = ck^2$ with $c = \frac{1}{2}$, rounded to an integer. Then $\frac{k(k-1)}{2n} = 1 + O\left(\frac{1}{k}\right)$ and $\frac{k^3}{n^2} = O\left(\frac{1}{k}\right)$, so

$$m(k) \leq m \leq 2^{k-1} \cdot \frac{k^2}{2} \cdot e^{1+O(1/k)} \ln 2 + 1 = (1 + o(1)) \frac{e \ln 2}{4} k^2 2^k \leq C k^2 2^k. \quad \blacksquare$$

Remark 1.1.16 (Estimating binomial coefficients). Exact computation of p is not needed for $O(k^2 2^k)$, only for the constant. The standard estimates

$$\begin{aligned} \frac{(n-k)^k}{k!} &\leq \binom{n}{k} = \frac{n(n-1)(n-2) \cdots (n-k+1)}{k!} \leq \frac{n^k}{k!}, \\ \left(\frac{n}{k}\right)^k &\leq \binom{n}{k} = \frac{n(n-1) \cdots (n-k+1)}{k(k-1) \cdots (k-k+1)} \leq \left(\frac{ne}{k}\right)^k \end{aligned}$$

follow from bounding each factor $n-i$ between $n-k$ and n , from $\frac{n-i}{k-i} \geq \frac{n}{k}$, and from $k! \geq (k/e)^k$. The first one already gives

$$p = \frac{2^{\binom{n/2}{k}}}{\binom{n}{k}} \geq 2 \cdot \frac{(n/2 - k)^k / k!}{n^k / k!} = \frac{1}{2^{k-1}} \left(1 - \frac{2k}{n}\right)^k \geq \frac{1}{2^{k-1}} e^{-4k^2/n} \quad \text{for } n \geq 4k,$$

using $1-x \geq e^{-2x}$ for $0 \leq x \leq \frac{1}{2}$. With $n = 4k^2$ this yields $m(k) \leq (1 + o(1)) 2e \ln 2 \cdot k^2 2^k$, the same order with a worse constant.

§1.1.vi. Better Lower Bounds: Random Greedy Coloring

Combining [Proposition 1.1.10](#) and [Theorem 1.1.15](#),

Theorem 1.1.17 (Erdős, 1963–64)
$$2^{k-1} \leq m(k) \leq Ck^2 2^k \text{ for some constant } C > 0.$$

Since then the lower bound has been improved by polynomial factors in k , while the upper bound $O(k^2 2^k)$ is still the best known.

Theorem 1.1.18 (Radhakrishnan–Srinivasan, 2000)
$$m(k) \geq c \sqrt{\frac{k}{\ln k}} 2^k \text{ for some constant } c > 0.$$
Theorem 1.1.19 (Pluhár, 2009)
$$m(k) \geq c k^{1/4} 2^k \text{ for some constant } c > 0.$$

We now prove [Theorem 1.1.19](#). The proof has two ideas: first a deterministic greedy coloring, whose failures we can describe exactly, and then a random order of the vertices, which makes such failures unlikely.

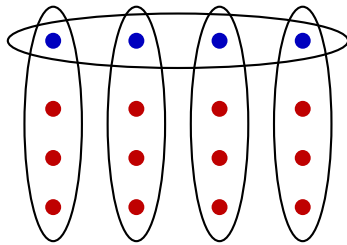
Proof. The proof of [Proposition 1.1.10](#) colored every vertex independently at random, which ignores the structure of H entirely. Instead:

Idea 1: Color the vertices greedily. Order the vertices $v_1, v_2, \dots, v_n$, and consider the vertices one by one. Color v_i red, unless that creates a monochromatic (red) edge, i.e. unless some edge contains v_i and all of its other vertices are already colored red, in which case we color v_i blue. Colors are never changed afterwards.

Observation .

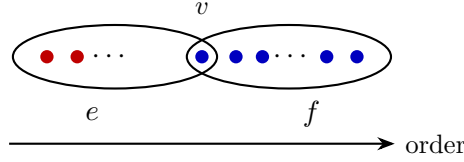
- (1) No monochromatic red edge: when the last vertex of an edge is reached and all the others are red, it is colored blue.
- (2) Monochromatic blue edges can occur. However, a vertex v is colored blue only because coloring it red would have completed a red edge. So every blue vertex v lies in some edge e such that all vertices of e other than v are red and come before v in the order.

For example, in the 4-uniform hypergraph below, order the twelve lower vertices first and the four top vertices last. The lower vertices are all colored red, since each vertical edge still has an uncolored top vertex. Each top vertex is then the last vertex of a vertical edge whose other vertices are red, so it is colored blue, and the horizontal edge ends up monochromatic blue.



Each blue vertex lies in a vertical edge that would otherwise be red.
The horizontal edge is monochromatic blue.

We say the greedy coloring *fails* if the coloring it produces is not a proper two-coloring, i.e. some edge is monochromatic. By (1), this happens exactly when some edge is monochromatic blue, as in the example. What must H and the order look like for this to happen? Suppose f is a monochromatic blue edge, and let v be the *first* vertex of f in the order. By (2), v lies in an edge e whose other vertices are red and come before v . Since all of f is blue, $e \cap f = \{v\}$; and since v is the last vertex of e and the first vertex of f , all of e comes before all of f :



We record what this argument shows. Note that it works for every order of the vertices.

Claim 1.1.1. For any order of the vertices, if the greedy coloring produces a monochromatic edge, then there are edges $e, f \in E(H)$ with $|e \cap f| = 1$ such that all of e comes before all of f in the order.

Here $e = f$ is possible only if $f = \{v\}$ is an edge of size 1: then $e = f = \{v\}$ and $|e \cap f| = 1$. Contrapositively, if no two edges meet in exactly one vertex, the greedy coloring cannot fail, whatever the order:

Corollary 1.1.20

If H is a hypergraph with $|e \cap f| \neq 1$ for all $e, f \in E(H)$, then H is two-colorable.

Proof. Fix any order $v_1, \dots, v_n$ of the vertices and run the greedy coloring. It colors every vertex red or blue, so it gives a two-coloring χ of V ; we show that no edge is monochromatic under χ .

- No edge is monochromatic red, by (1) of the observation.
- Suppose some edge f is monochromatic blue. By Claim 1.1.1 there would be edges e, f with $|e \cap f| = 1$, contradicting the assumption on H .

So every edge contains both colors, and H is two-colorable. ■

Idea 2: Order the vertices uniformly at random. Let H be a k -uniform hypergraph with m edges. Choose an order σ of V uniformly at random among all $n!$ orders, run the greedy coloring along σ , and let χ_σ be the resulting two-coloring. We will show that

$$\mathbb{P}(\chi_\sigma \text{ is not a proper two-coloring}) < 1 \quad \text{whenever } m < c k^{1/4} 2^k.$$

Then some order σ makes χ_σ proper, so H is two-colorable, and hence $m(k) \geq c k^{1/4} 2^k$.

Write $e \prec f$ if all of e comes before all of f in σ . By (1), χ_σ is not proper exactly when it has a monochromatic blue edge, and by Claim 1.1.1 this forces a pair $e \prec f$ with $|e \cap f| = 1$. Hence, by the union bound,

$$\begin{aligned} \mathbb{P}(\chi_\sigma \text{ is not proper}) &= \mathbb{P}(\exists \text{ monochromatic blue edge}) \\ &\leq \mathbb{P}(\exists e, f \in E(H) : |e \cap f| = 1, e \prec f) \\ &\leq \sum_{\substack{(e,f) \\ |e \cap f| = 1}} \mathbb{P}(e \prec f). \end{aligned}$$

Fix an ordered pair (e, f) with $|e \cap f| = 1$, so $|e \cup f| = 2k - 1$. Since σ is uniform, the order it induces on $e \cup f$ is uniform among all $(2k - 1)!$ orders of $e \cup f$. In an order with $e \prec f$, the $k - 1$ vertices of $e \setminus f$ come first, in any of $(k - 1)!$ orders, then the common vertex, then the

$k - 1$ vertices of $f \setminus e$, in any of $(k - 1)!$ orders. Thus

$$\mathbb{P}(e \prec f) = \frac{\overbrace{(k-1)!}^{\text{ordering } e \setminus f \text{ at front}} \overbrace{(k-1)!}^{\text{ordering } f \setminus e \text{ at back}}}{\underbrace{(2k-1)!}_{\text{total \# of orderings of } e \cup f}}.$$

There are at most m^2 ordered pairs (e, f) , so

$$\mathbb{P}(\chi_\sigma \text{ is not proper}) \leq m^2 \frac{(k-1)!(k-1)!}{(2k-1)!}.$$

By Stirling's approximation $N! \sim \sqrt{2\pi N}(N/e)^N$, in the form $\binom{2N}{N} \sim \frac{4^N}{\sqrt{\pi N}}$ from the proof of [Proposition 1.1.12](#),

$$m^2 \frac{(k-1)!(k-1)!}{(2k-1)!} = \frac{m^2}{(2k-1)\binom{2k-2}{k-1}} \sim \frac{m^2 \sqrt{\pi k}}{2k \cdot 4^{k-1}} = \frac{2\sqrt{\pi} m^2}{\sqrt{k} 2^{2k}},$$

which is < 1 if $m^2 < c'\sqrt{k} 2^{2k}$, i.e. if $m < c k^{1/4} 2^k$ (any $c < (2\sqrt{\pi})^{-1/2}$ works for large k). This is what we wanted to show. ■

Theorem 1.1.21 (Cherkashin–Kozik, 2014)

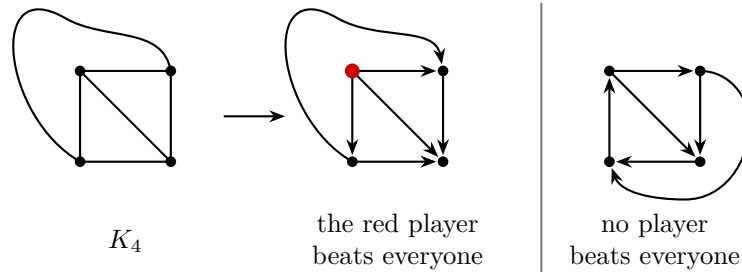
$$m(k) \geq c \sqrt{\frac{k}{\ln k}} 2^k \text{ for some constant } c > 0.$$

Proof idea. Exactly the same random greedy coloring of Pluhár, analyzed a little better. This recovers the bound of Radhakrishnan–Srinivasan with a much simpler proof. ■

§1.2. Schütte Tournaments

Definition 1.2.1 (Tournament). A **tournament** is an orientation of the edges of the complete graph K_n . **Orientation** means: for each edge $\{u, v\}$, we choose to orient the edge either from u to v , giving the arc (u, v) , or from v to u , giving the arc (v, u) . We say u **beats** v if $(u, v) \in E(T)$.

Think of a round-robin tournament: every two of the n players play one game, and the arc points from the winner to the loser. Sometimes the results determine a winner, a player who beats everyone else, and sometimes they do not.

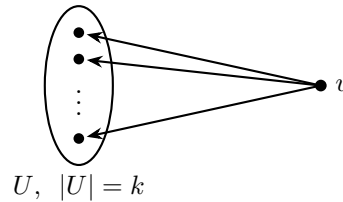


In the right tournament, the top-left player beats two others but loses to the bottom-left one, who in turn loses to both right players; every player loses to someone, so this condition cannot decide a winner. A winner is a single vertex u s.t. every other vertex is beaten by u . When there is no winner, we can relax the requirement: instead of one player who beats everyone, look for a small group of k players who together beat everyone else. This leads to the following question.

Question 1.2.2. Given a tournament T on n vertices, can we find a set U of k vertices s.t. for every vertex $v \notin U$, $\exists u \in U$ that beats v , i.e. $(u, v) \in E(T)$?

For $k = 1$ this asks for a winner. Schütte asked whether for every k there are tournaments in which no k players jointly beat everyone else.

Definition 1.2.3 (Schütte property). We say a tournament has the **k -Schütte property** if $\nexists$ such a set of k vertices, i.e. for any set U of k vertices, $\exists v \notin U$ s.t. v beats every $u \in U$.



Example 1.2.4. For $k = 1$, the 1-Schütte property says that there is no winner: every vertex is beaten by some other vertex. The right tournament above has it, and so does the cyclic triangle $a \rightarrow b \rightarrow c \rightarrow a$. No tournament on $n \leq 2$ vertices has it, so 3 is the fewest vertices for $k = 1$.

Question 1.2.5. How small can a tournament with the k -Schütte property be?